


```

barplot(matrix(answers),

main = "Distribution of Question 1 Answers",

xlab = "Question 1",

ylab = "Number of Responses",

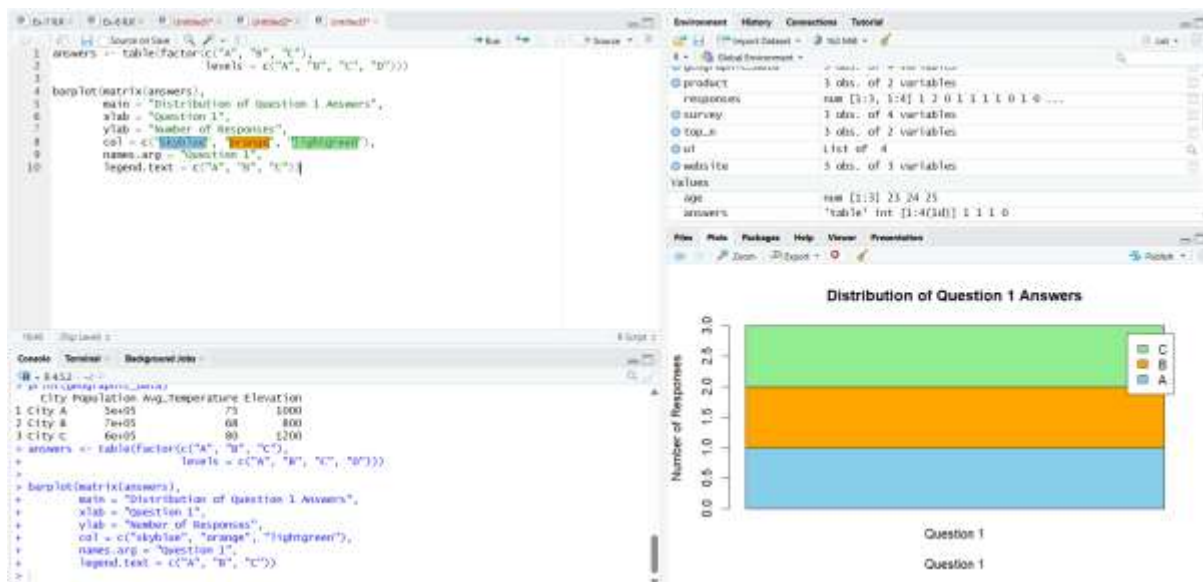
col = c("skyblue", "orange", "lightgreen"),

names.arg = "Question 1",

legend.text = c("A", "B", "C"))

```

Output



A stacked bar chart displaying the distribution of A, B and C answers for Question 1.

Observation

Question 1 contains one response each for A, B and C.

2. Radar Chart – Overall Response Pattern

Aim

To create a radar chart to visualize the overall response pattern across all three questions.

R Code

R

```
answers <- matrix(c(

  1, 1, 1, 0,

  2, 1, 0, 0,

  1, 1, 1, 1

), nrow = 3, byrow = TRUE)

colnames(answers) <- c("A", "B", "C", "D")

rownames(answers) <- c("Question 1", "Question 2", "Question 3")

angles <- seq(0, 2*pi, length.out = 5)

plot(0, 0,

     type = "n",

     xlim = c(-2, 2),

     ylim = c(-2, 2),

     axes = FALSE,

     xlab = "",

     ylab = "",

     main = "Overall Survey Response Pattern")

for(i in 1:3) {

  values <- c(answers[i, ], answers[i, 1])

  x <- values * cos(angles)

  y <- values * sin(angles)

  lines(x, y)

  points(x, y)
```

```
}
```

```
text(1.5, 0, "A")
```

```
text(0, 1.5, "B")
```

```
text(-1.5, 0, "C")
```

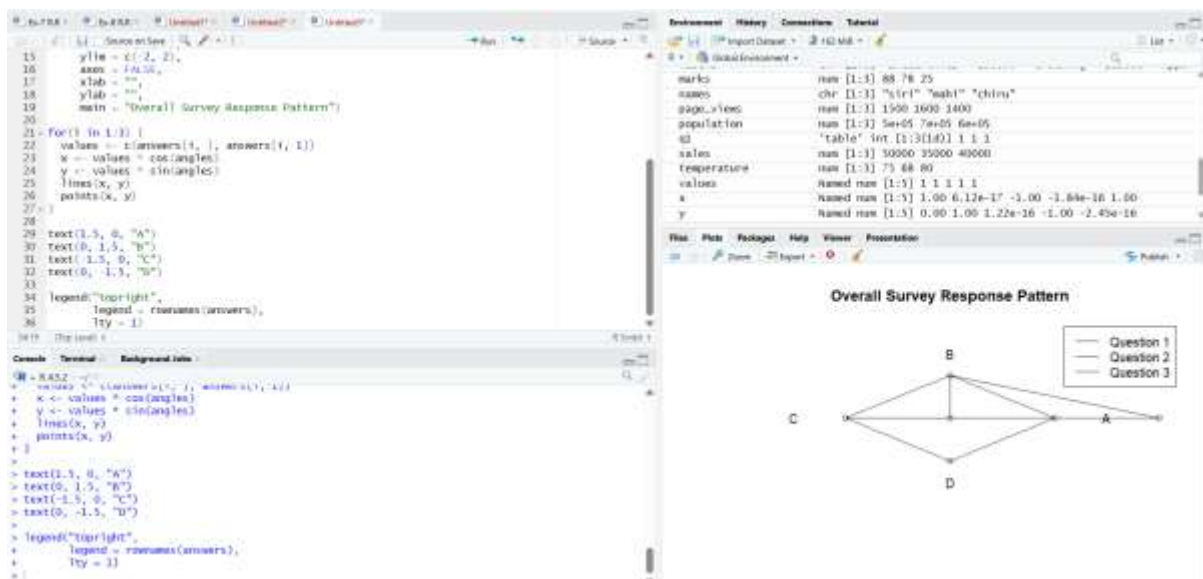
```
text(0, -1.5, "D")
```

```
legend("topright",
```

```
  legend = rownames(answers),
```

```
  lty = 1)
```

Output



A radar-style chart showing the response pattern for Question 1, Question 2 and Question 3.

Observation

The radar chart helps compare the distribution of answers across all three questions. Answer

3. Survey Response Table

Aim

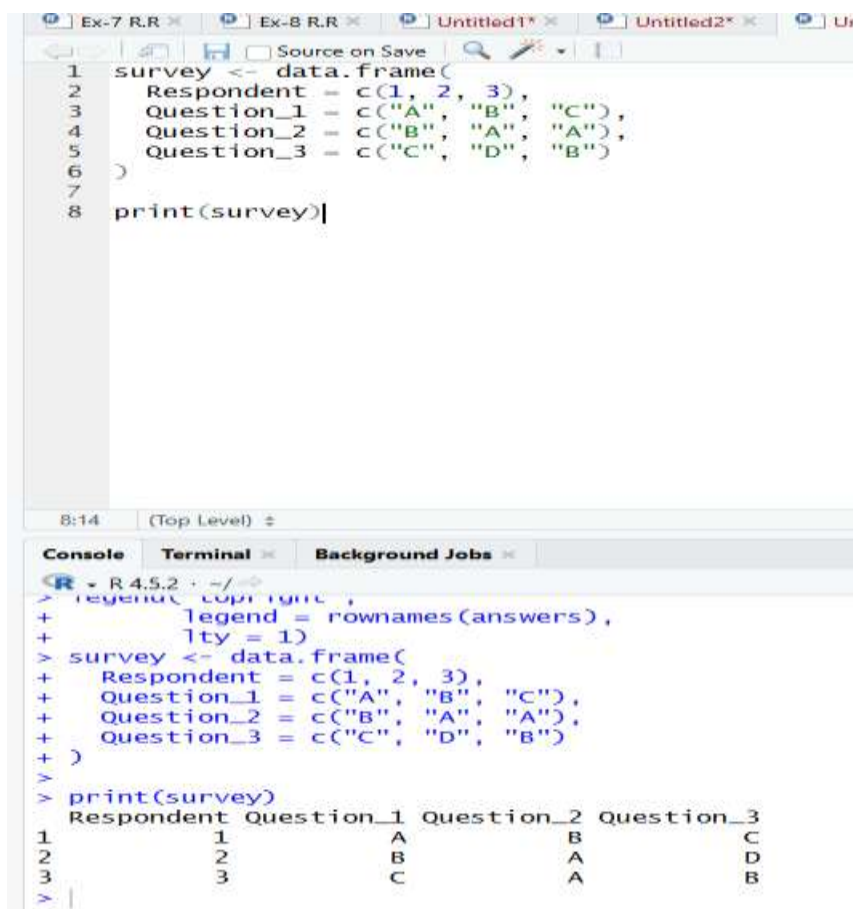
To create a table displaying the survey response data for each respondent.

R Code

R

```
survey <- data.frame(  
  
  Respondent = c(1, 2, 3),  
  
  Question_1 = c("A", "B", "C"),  
  
  Question_2 = c("B", "A", "A"),  
  
  Question_3 = c("C", "D", "B")  
  
)  
  
print(survey)
```

Output



The screenshot shows an R console window with the following code and output:

```
> legend("topleft",  
+       legend = rownames(answers),  
+       lty = 1)  
> survey <- data.frame(  
+   Respondent = c(1, 2, 3),  
+   Question_1 = c("A", "B", "C"),  
+   Question_2 = c("B", "A", "A"),  
+   Question_3 = c("C", "D", "B")  
+ )  
> print(survey)
```

	Respondent	Question_1	Question_2	Question_3
1	1	A	B	C
2	2	B	A	D
3	3	C	A	B

Observation

The table clearly displays the answers given by each respondent for all three questions.

4. Tableau Dashboard

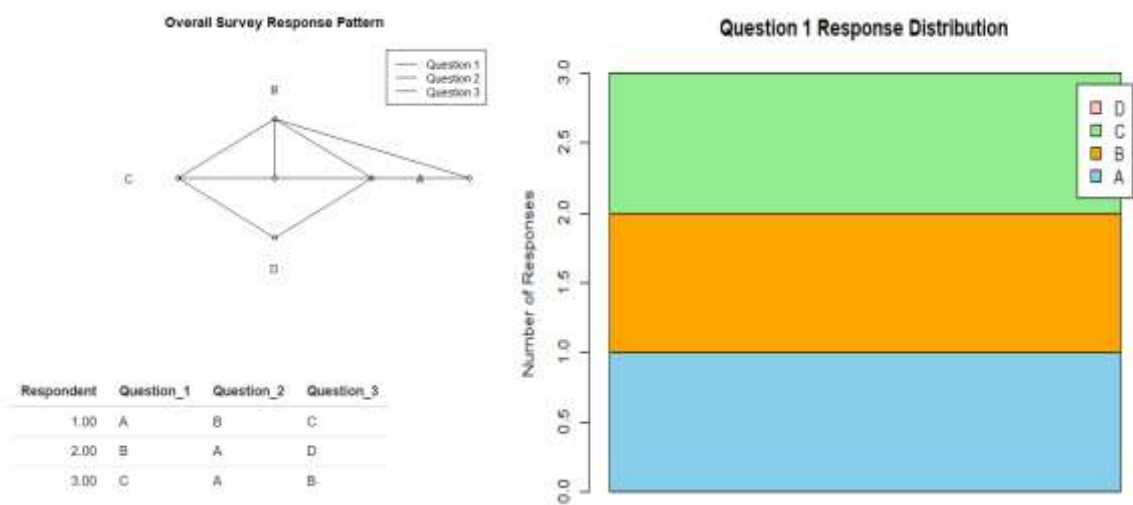
Aim

To develop an interactive dashboard combining the stacked bar chart, radar chart and survey response table.

Tableau Procedure

1. Open Tableau.
2. Import the Survey Results dataset.
3. Create a Stacked Bar Chart for Question 1.
4. Create a Radar Chart to display the overall response pattern.
5. Create a Table containing Respondent, Question 1, Question 2 and Question 3.
6. Select Dashboard → New Dashboard.
7. Drag the stacked bar chart into the dashboard.
8. Drag the radar chart into the dashboard.
9. Drag the survey table into the dashboard.
10. Add suitable titles and labels.
11. Arrange the charts neatly.
12. Save the dashboard.

Output:



Result

The survey responses were successfully analyzed using a stacked bar chart, radar chart and table. The dashboard provides an interactive view of the response patterns across all three questions.